

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. This document records the testing approach and available results for EduGenie.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API request/response testing)
Type of Testing	Load-oriented API testing; stress testing after baseline validation.
Target Module / API	/qa, /explain, /quiz, /summarize, /learn/recommendations
Test Environment	Local FastAPI/Uvicorn environment
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Q&A endpoint request handling	Planned: 1	Planned	Valid request returns without server error.
2	Quiz generation endpoint	Planned: 1	Planned	Expected quiz structure is returned.
3	Summary with long text	Planned: 1	Planned	Summary returns without request failure.
4	Learning recommendations endpoint	Planned: 1	Planned	Recommendation response returns successfully.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Pending execution	No measured load result supplied.
2	Response Time (Max)	< 5 seconds	Not measured	Pending execution	No measured load result supplied.
3	Throughput (Req/sec)	Baseline	Not measured	Pending execution	Requires concurrent requests.
4	Error Rate	< 1%	Not measured	Pending execution	Requires load-test run.
5	CPU Utilization	< 80%	Not measured	Pending execution	Requires monitoring during load.
6	Memory Utilization	< 80%	Not measured	Pending execution	Requires monitoring during load.

Step 4: Observations & Analysis

Key Findings:

EduGenie exposes FastAPI endpoints for multiple learning tasks.
Baseline performance values must be collected by repeated requests.

Bottlenecks Identified:

No measured bottleneck is claimed without an executed performance test.

Optimization Steps Taken:

Task-specific modules and a shared Gemini client configuration are used.

Step 5: Screenshots / Evidence

Attach actual performance-tool screenshots below. No fabricated evidence is inserted.

